Bálint Tasnádi

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Education

Budapest University of Technology and Economics (BME) (2024–)
Physicist-engineer BSc, Specialization: Nuclear Technology and Sustainable Energetics
5th semester, GPA: 4.81/5

Eötvös József Gimnázium, Tata (2020–2024)

Experience & Projects

Franck-Hertz Experiment Hardware (2026)
- Designed and manufactured 6 sets of custom laboratory equipment for undergraduate laboratories at the BME Institute of Atomic Physics.
- Developed a custom analog electrode driver, an ultra-low bias (LMC661) transimpedance amplifier for measuring pA tube currents, and a solid-state PID thermostat.
- Conducted several laboratory sessions using the manufactured devices.

Cryogenic Sample Holder System (2025)
- Designed custom PCBs (motherboard and daughterboards) for testing quantum electronic samples in 3He-4He dilution fridges for the BME Nanoelectronics Research Group.
- Implemented high-density Samtec connectors and gold-plated contacts to allow for direct sample wire bonding.

Advanced Test Equipment Repair (2022–2026)
- Performed component-level diagnostics and repair of over a dozen high-end instruments, including a 6.5-digit Keithley 2000 DMM, Tektronix analog storage oscilloscopes, and sensitive picoammeters.
- Executed complex repairs including severe PCB corrosion mitigation, PLL clock signal restoration, analog circuit troubleshooting

Independent Hardware Projects
- **Nixie-tube 5.5-digit DMM:** Designing a benchtop voltmeter integrating high-voltage shift registers, IN-12B/IN-15A tubes, and an ADS1259 24-bit ADC.
- **Gamma Spectrometer:** Assembled and calibrated a NaI(Tl) scintillation detector utilizing a SiPM and RP2040 for isotopic spectra analysis.
- **Power Electronics & Mods:** Built an IGBT H-bridge solid-state Tesla coil and designed custom 555-timer PWM analog circuitry to convert vintage iMacs into standalone monitors.

Skills

- **Hardware:** Analog & Digital Circuit Design, PCB Layout, SMD & THT Soldering, Component-Level Troubleshooting, Precision Metrology & Test Equipment
- **Software & Simulation:** Python, KiCAD, LTspice, C/C++, C#, MATLAB, MS Office
- **Languages & Other:** English (C1 Proficient, full-English BSc program), Hungarian (Native), Technical Writing

Awards, Publications & Conferences

- **23rd National Scientific Students' Associations Conference (OTDK)** in Physics, Mathematics, Earth Sciences: Special Prize in Classical Physics (2023)
- **National Secondary School Academic Competition (OKTV)** in Physics: 12th Place (2023)
- **ELTE TTK TDK**: First Prize (High School Section, 2022)
- **"Atomoktól a csillagokig" Talk Series**: Presentation on the Marangoni effect (2023)
- **Publication**: B. Tasnádi, K. Rossz, Á. Szeidemann, "Marangoni-droplet explosion investigation in high school" (*Hungarian Physical Review*, April 2026)
- **VI. European Conference of RM@Schools**: "Experiments with critical raw material tin" (2021)
- **2022 Researchers' Night**: "High voltage in the service of science" (Eötvös József Gimnázium, Tata)